

State of Document AI

A Comprehensive Survey of Ingestion Pipelines

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Published: June 2026

1. Introduction

Document Artificial Intelligence represents a milestone in intelligent process automation. A typical pipeline comprises document parsing, OCR (Optical Character Recognition) for scanned pages, document classification, chunking, database storage, and retrieval-augmented generation. Recent trends focus heavily on data security at every layer, especially when processing sensitive documents.

2. Key Metrics

In our assessment of free OCR libraries, Tesseract achieves 92.5% character accuracy on clean documents. However, for tables and complex layouts, advanced multimodal models like Gemini 2.5 Flash deliver up to 99.1% accuracy by retaining row-column spatial structures. Security encryption at rest adds negligible latency (under 5 milliseconds per document) while completely mitigating physical theft risk.